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EXPERIMENT 5
Word Stemming using Porter Stemmer

Aim

To perform word stemming on a list of words using the Porter Stemmer algorithm in Python.

Algorithm

• Import PorterStemmer from the nltk.stem module.

• Create an object of the PorterStemmer class.

• Store the list of words in a list.

• Use a loop to process each word.

• Apply the stem() method to each word to obtain its stem.

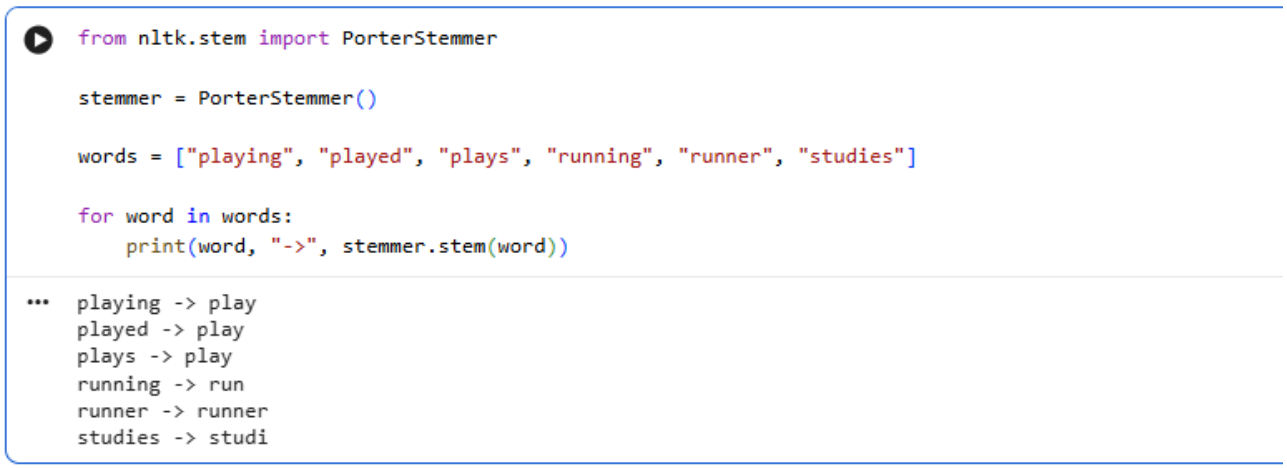
• Display each original word and its stem.

Code

```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

words = ["playing", "played", "plays", "running", "runner", "studies"]

for word in words:
    print(word, "->", stemmer.stem(word))
```

Output

Output screenshot provided:



Result

The Porter Stemmer algorithm was applied successfully to the given list of words, producing stems such as “playing” → “play”, “running” → “run”, and “studies” → “studi”.